

Setup

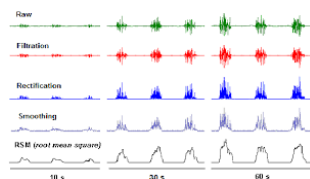
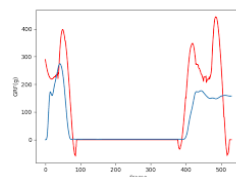
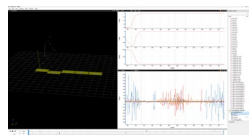
Models



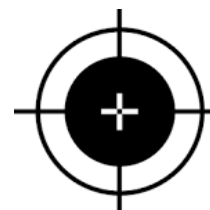
MRI_scaled_BG
(rater 1)

MRI_scaled_KS
(rater 2)

Generic_scaled



Tasks



Calibration

CEINMS with walking data – New cfg settings

calibration with lower penalty for normalisedFibreLength = 10 (instead of 1000)

<parametersToCalibrate>

<parameter name="c1">-0.99 -0.05</parameter>

<parameter name="c2">-0.95 -0.05</parameter>

<parameter name="shapefactor">-2.999 -0.001</parameter>

<parameter name="optimalFiberLength">0.5 1.5</parameter> (from 0.95 to 1.05)

<parameter name="tendonSlackLength">0.5 1.5</parameter>

<parameter name="strengthCoefficient">0.75 3.5</parameter>

WORKED -> increases in TSL, increased OFL, reduced moment error by 1/5

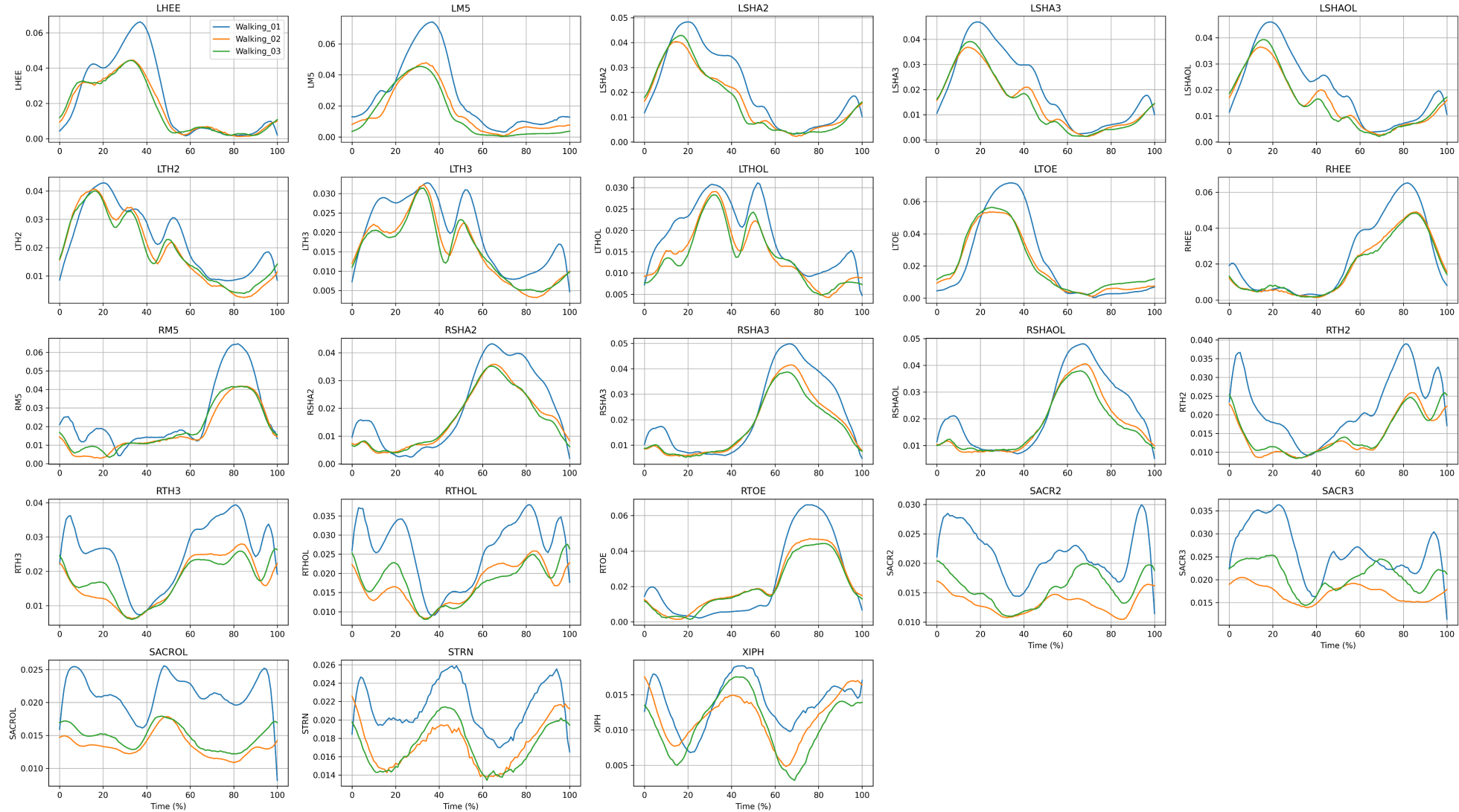
OPTIMISER RUNNING FOR Squat_bar_01 but stuck in walking (calibration was on walking)

CEINMS EXECUTABLE rung

Athlete 03 – session 2 - Walking

Scaled_model

Kinematic
errors



All bellow 0.07
(max)

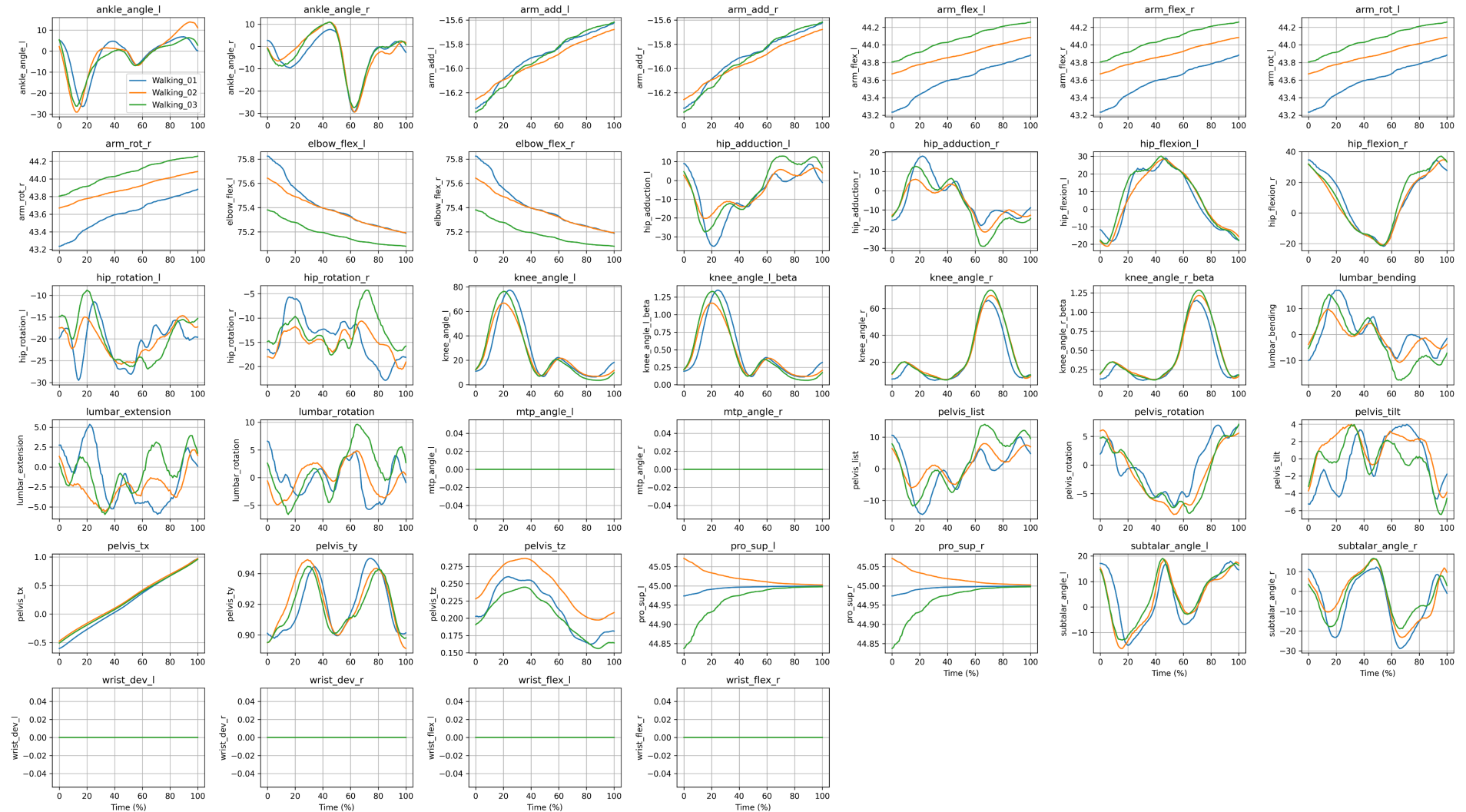
Athlete 03 – session 2 - Walking

Scaled_model



Kinematics

Good



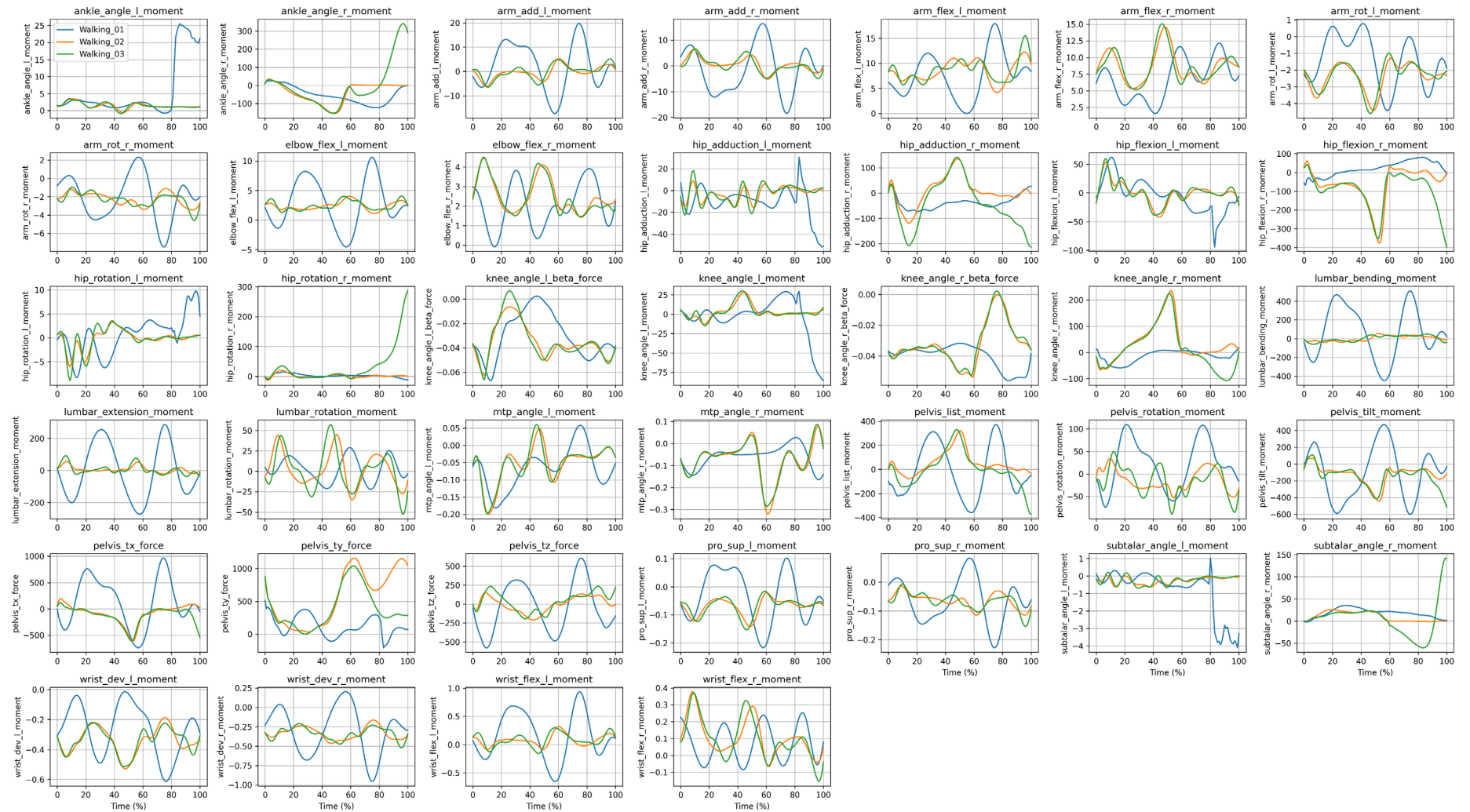
Athlete 03 – session 2 – Walking - PROBLEM

Scaled_model



Moments

Errors



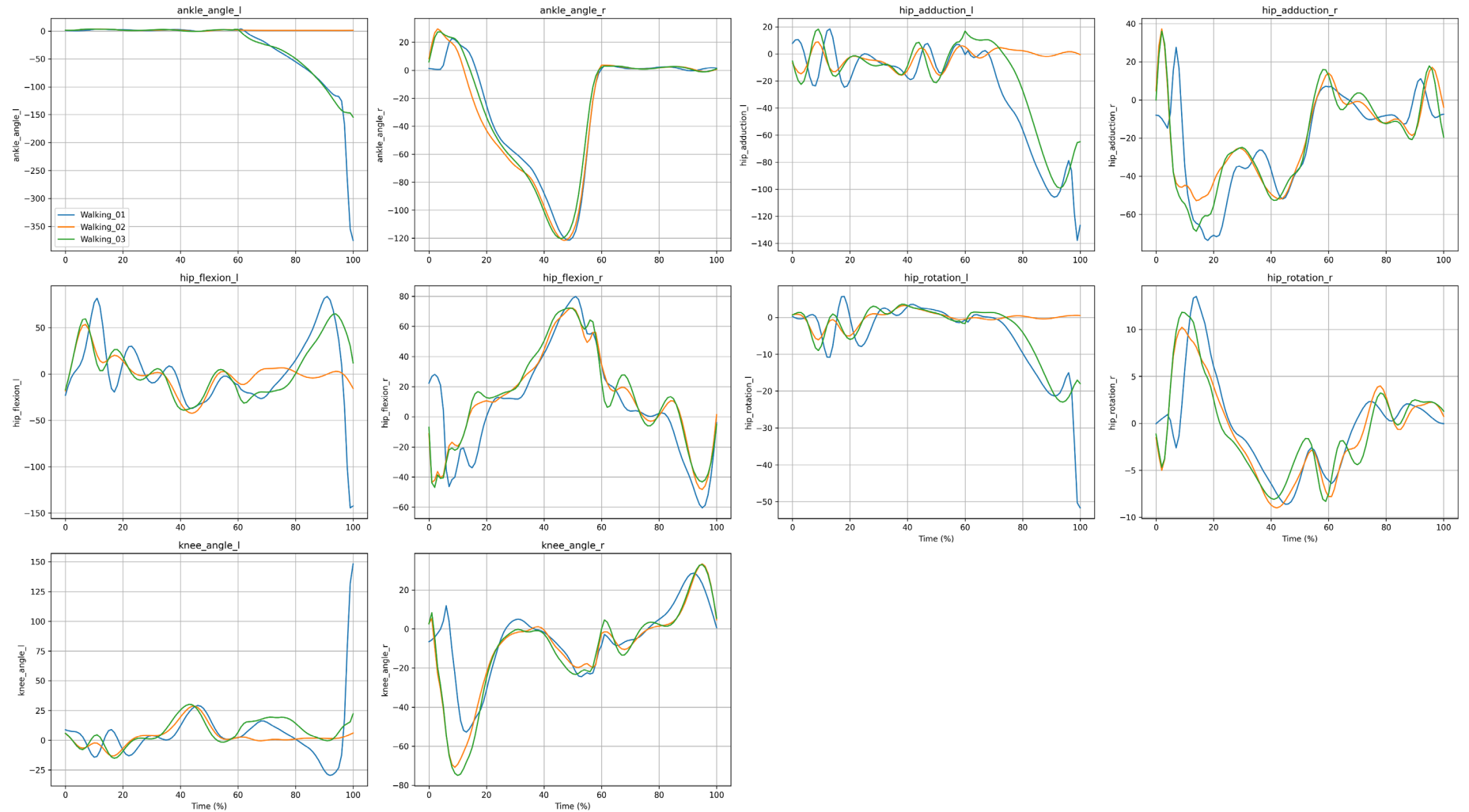
Athlete 03 – session 2 - Walking



Scaled_model

Moments

Fixed



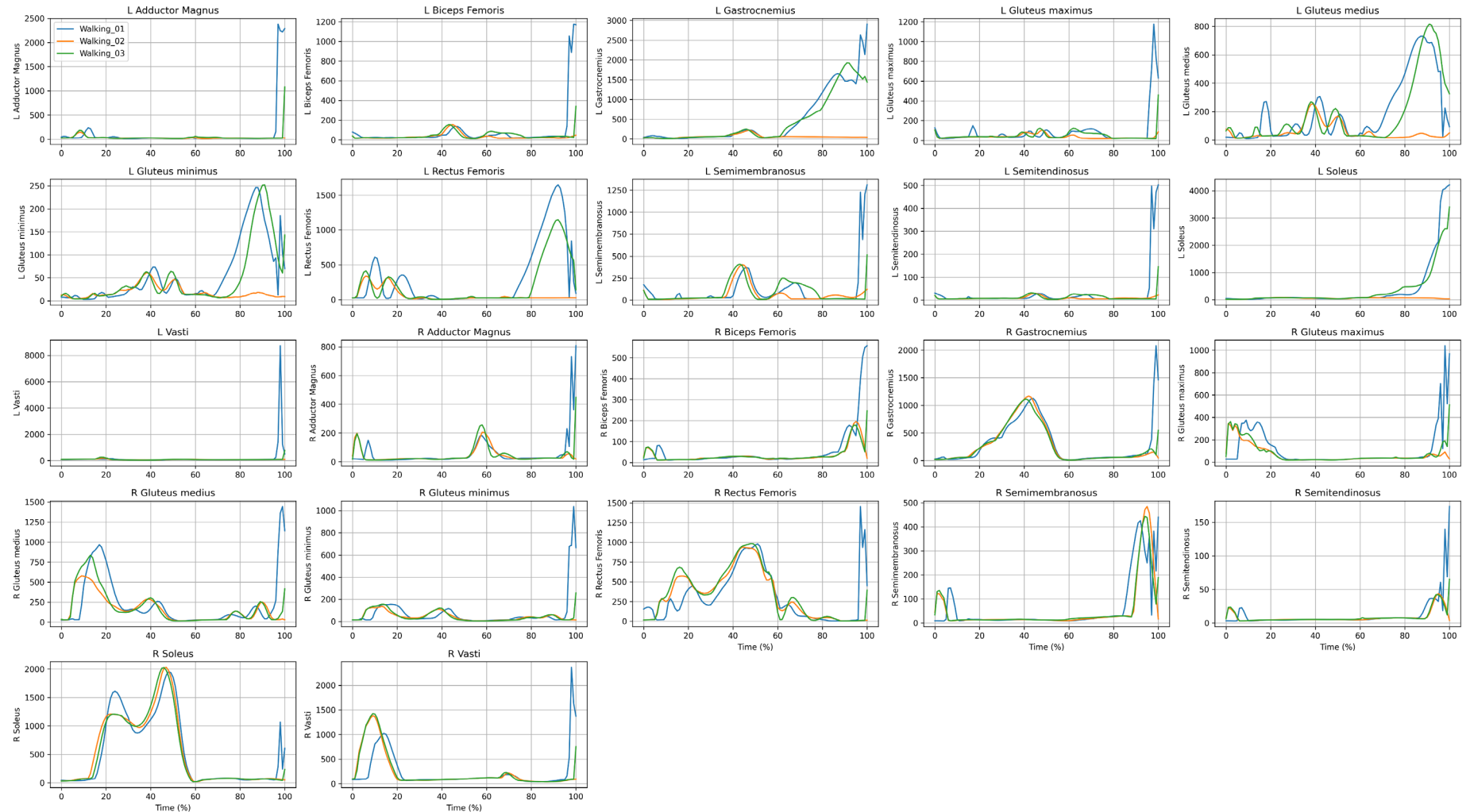
Athlete 03 – session 2 - Walking

Scaled_model

Forces SO



Forces SO look good



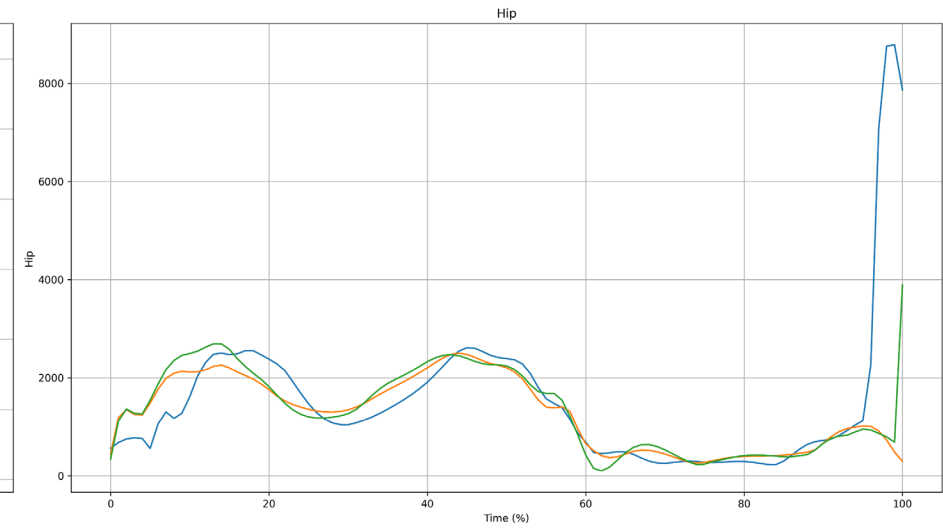
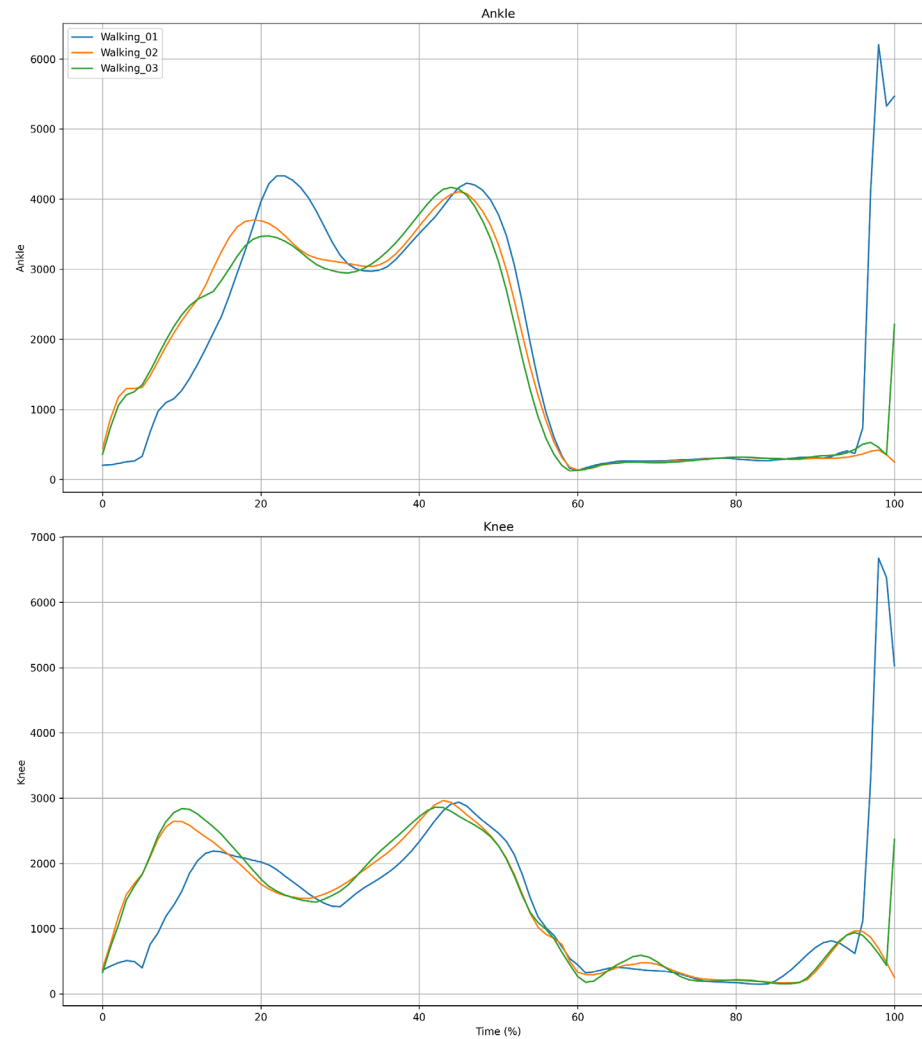
Athlete 03 – session 2 - Walking



Scaled_model

JRA SO

Not too bad



Athlete 03 – session 2 - Walking



Scaled_model

CEINMS
Calibration

Low error
moments

```
Iteration 44:
Total loss = 411.258 (best: 335.294, iterations without improvement: 20)
- Moment error = 146.892
- Normalised fiber length penalty = 253.601
- Tendon strain penalty = 2.97666
- Synthesised excitations squared = 0.033511
- Synergy extraction = 1.44306
- EMG tracking error = 0.865686
- Primitives range penalty = 0.15814
- Muscle weightings range penalty = 0.41923
- Synthesised excitations range penalty = 0
- Parameters penalty = 6.31235
Model accuracy:
- Trial inputData:
- hip_flexion_r: RMSE = 10.17 Nm, R2 = 0.933872
- hip_adduction_r: RMSE = 2.28588 Nm, R2 = 0.993355
- hip_rotation_r: RMSE = 4.32933 Nm, R2 = 0.498174
- knee_angle_r: RMSE = 24.607 Nm, R2 = 0.0402042
- ankle_angle_r: RMSE = 21.744 Nm, R2 = 0.792139
- hip_flexion_l: RMSE = 15.86 Nm, R2 = 0.77173
- hip_adduction_l: RMSE = 5.55019 Nm, R2 = 0.886378
- hip_rotation_l: RMSE = 2.30195 Nm, R2 = 0.703603
- knee_angle_l: RMSE = 51.8587 Nm, R2 = 0.0828619
- ankle_angle_l: RMSE = 8.18479 Nm, R2 = 0.458428
Synergy accuracy:
- Mean RMSE = 0.0122868
- Mean R2 = 0.593664
=====

Excitation synthesizer accuracy:
- Mean RMSE = 0.0333739
- Mean R2 = 0.486322

Calibration finished
Time taken: 15 seconds
```

Athlete 03 – session 2 - Walking

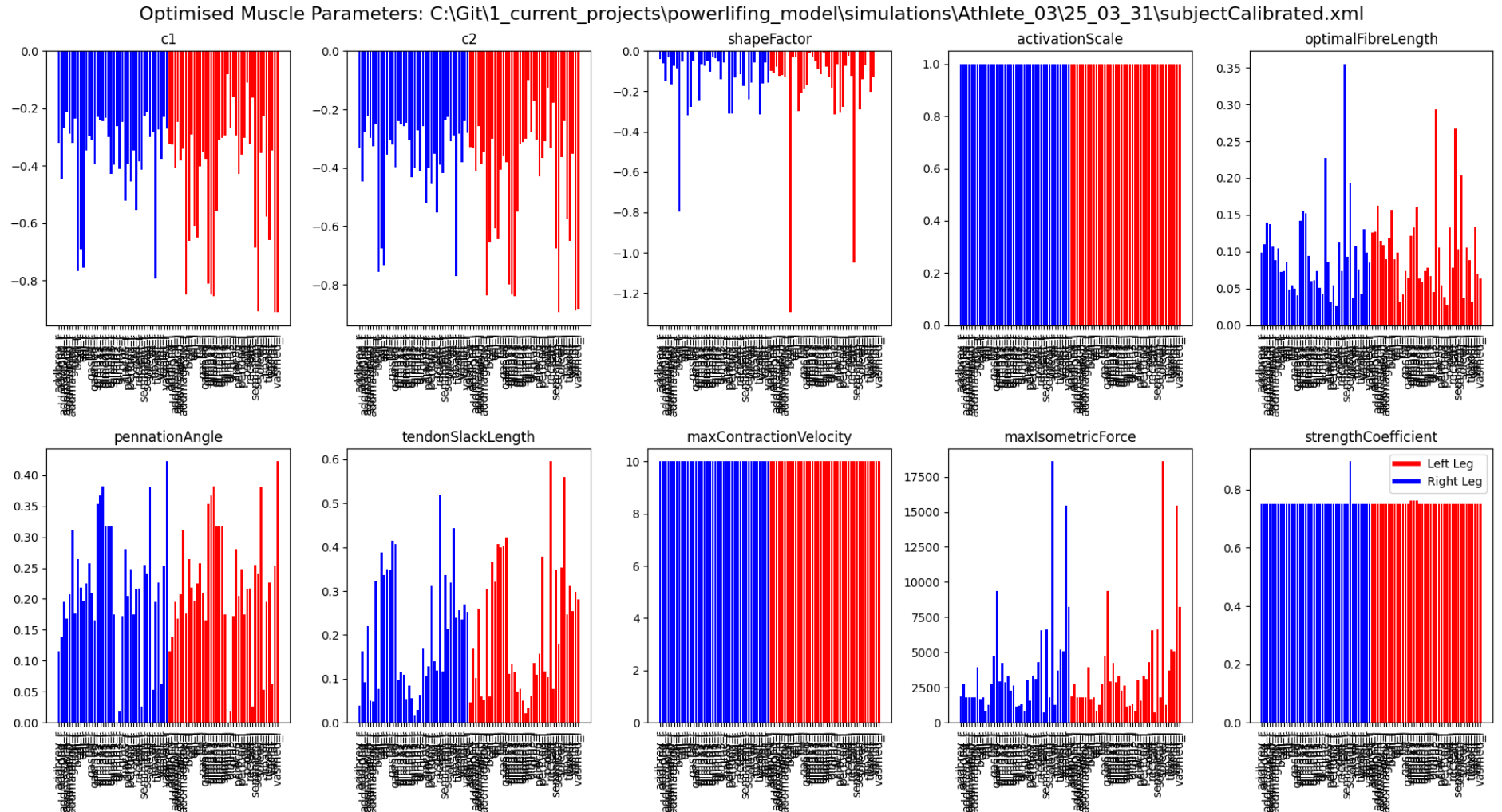
Possible due to
EMG signals
being too high



Scaled_model

CEINMS
Calibration

Problem with
strength
coefficient



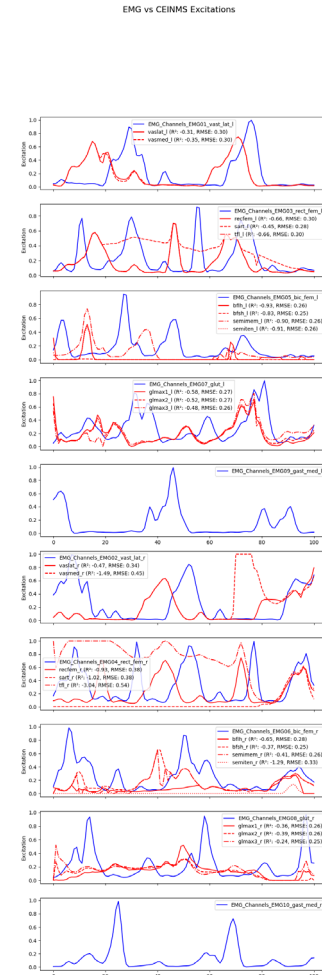
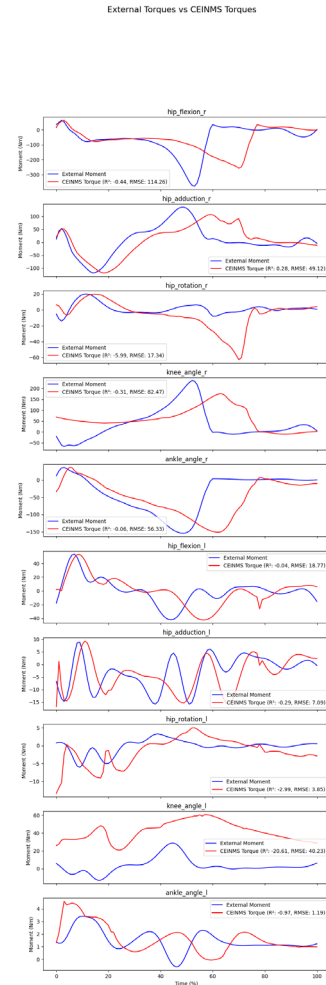
Athlete 03 – session 2 - Walking

Scaled_model

CEINMS
Calibration

CEINMS
Execution
Walking 02

High errors
One trial did not
converge



Athlete 03 – session 2 - Walking

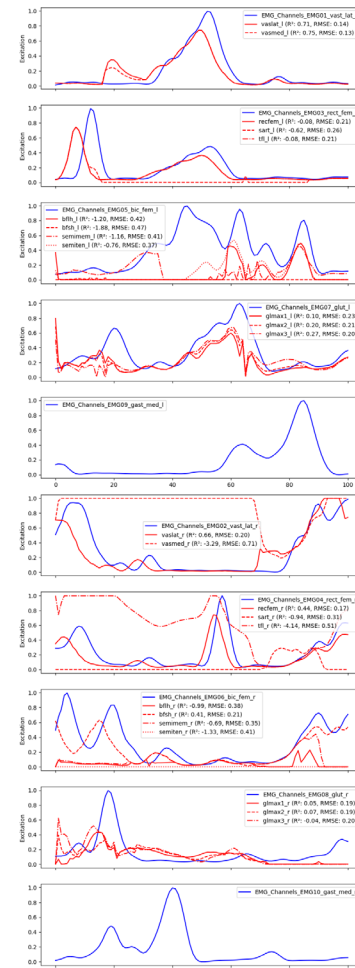
Scaled_model

CEINMS
Calibration

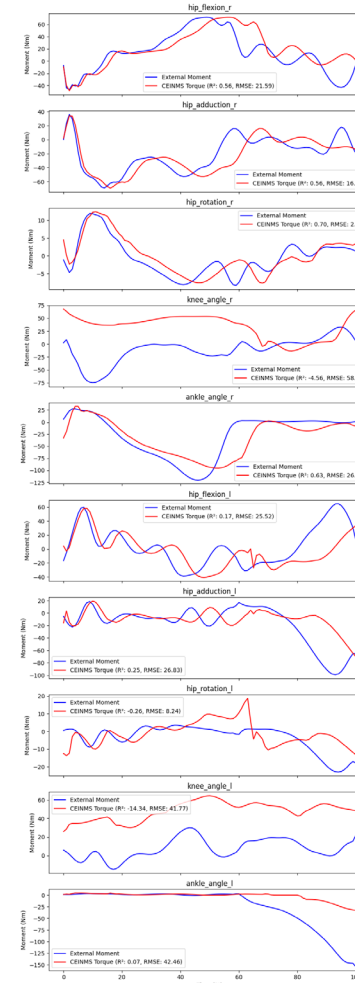
CEINMS
Execution
Walking 03

High errors
One trial did not
converge

EMG vs CEINMS Excitations



External Torques vs CEINMS Torques



Athlete 03 – session 2 - Walking

Scaled_model



CEINMS
Calibration



CEINMS
Execution
Walking 03

Have you tired?
- Re run calibration with emg
signal amplitude x 0.7
(mimic static opt)



Athlete 03 – session 2 - Walking

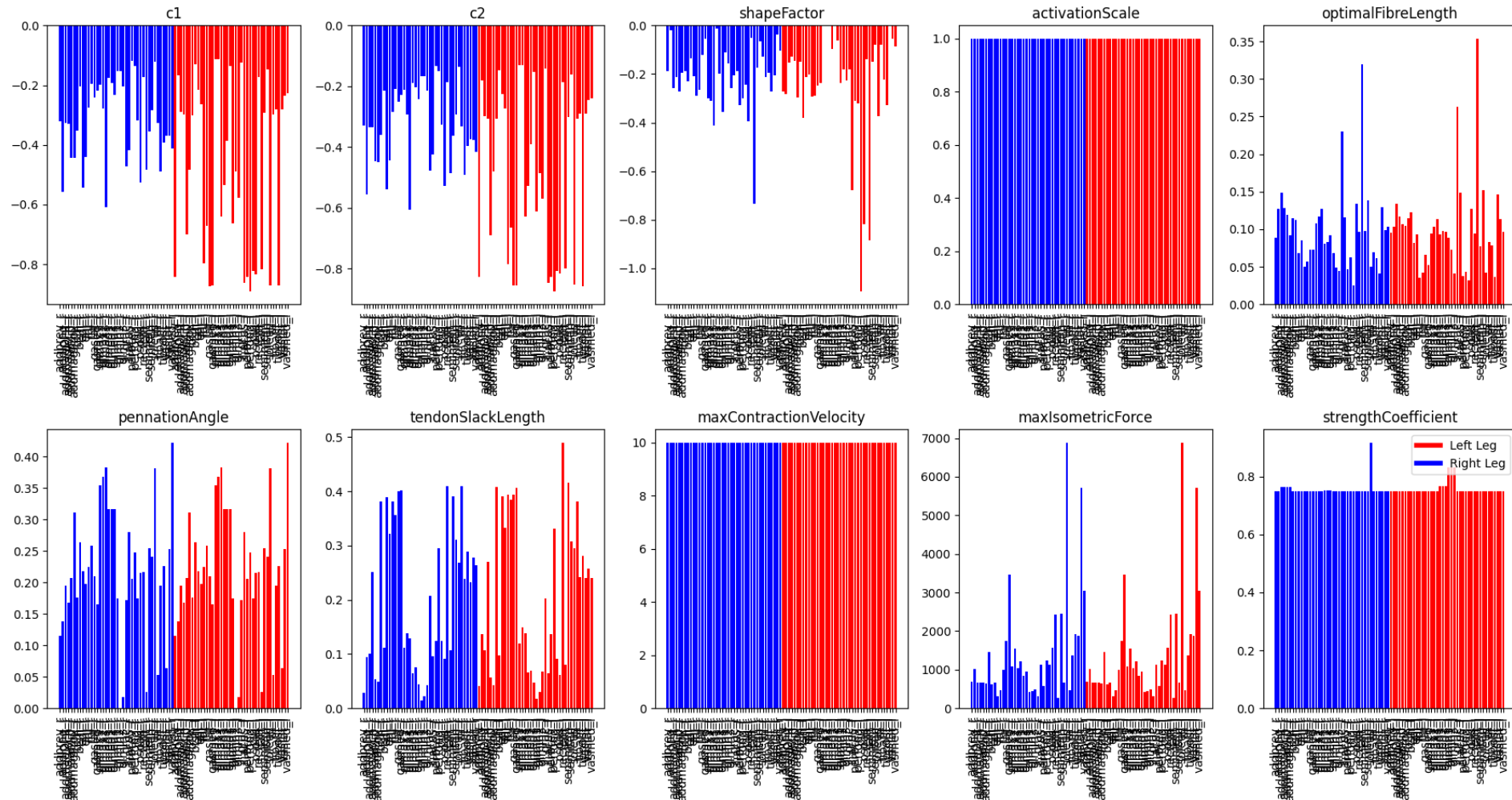


Scaled_model

CEINMS
Calibration
EMG amplitude x0.7

Still same
strength
problem

Optimised Muscle Parameters: ..\subjectCalibrated.xml



Athlete 03 – session 2 - Walking

Scaled_model



CEINMS
Calibration
EMG amplitude x0.7

Moment errors
slightly
increased

Iteration 40:
Total loss = 247.32 (best: 224.991, iterations without impro
- Moment error = 195.316
- Normalised fiber length penalty = 36.2509
- Tendon strain penalty = 3.05449
- Synthesised excitations squared = 0.0598523
- Synergy extraction = 2.76369
- EMG tracking error = 2.22977
- Primitives range penalty = 0.104212
- Muscle weightings range penalty = 0.42509
- Synthesised excitations range penalty = 0.00461361
- Parameters penalty = 9.87488

Model accuracy:

- Trial inputData:
- hip_flexion_r: RMSE = 18.2296 Nm, R2 = 0.690471
- hip_adduction_r: RMSE = 7.68634 Nm, R2 = 0.883076
- hip_rotation_r: RMSE = 3.58225 Nm, R2 = 0.701399
- knee_angle_r: RMSE = 15.5718 Nm, R2 = 0.495419
- ankle_angle_r: RMSE = 19.9895 Nm, R2 = 0.806126
- hip_flexion_l: RMSE = 46.8802 Nm, R2 = 0.147181
- hip_adduction_l: RMSE = 28.8558 Nm, R2 = 0.494767
- hip_rotation_l: RMSE = 9.54536 Nm, R2 = 0.418304
- knee_angle_l: RMSE = 9.93826 Nm, R2 = 0.449804
- ankle_angle_l: RMSE = 35.0373 Nm, R2 = 0.226098

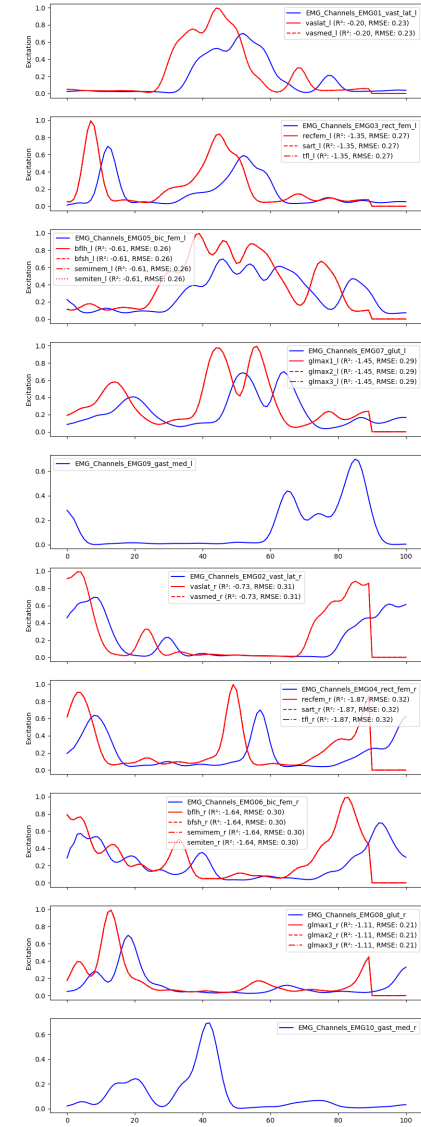
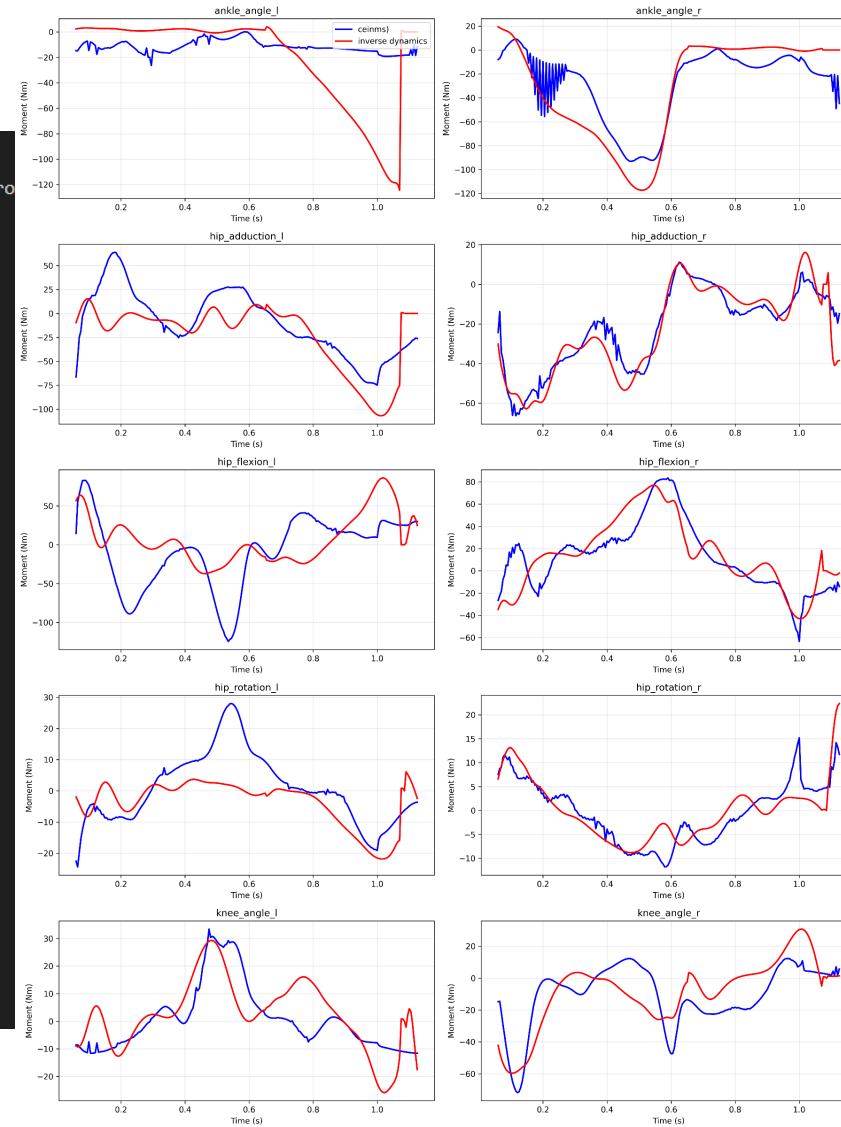
Synergy accuracy:

- Mean RMSE = 0.0266871
- Mean R2 = 0.987508

Excitation synthesizer accuracy:

- Mean RMSE = 0.0206196
- Mean R2 = 0.988462

DOF Angles and Moments Comparison





Scaled_model

CEINMS
Calibration

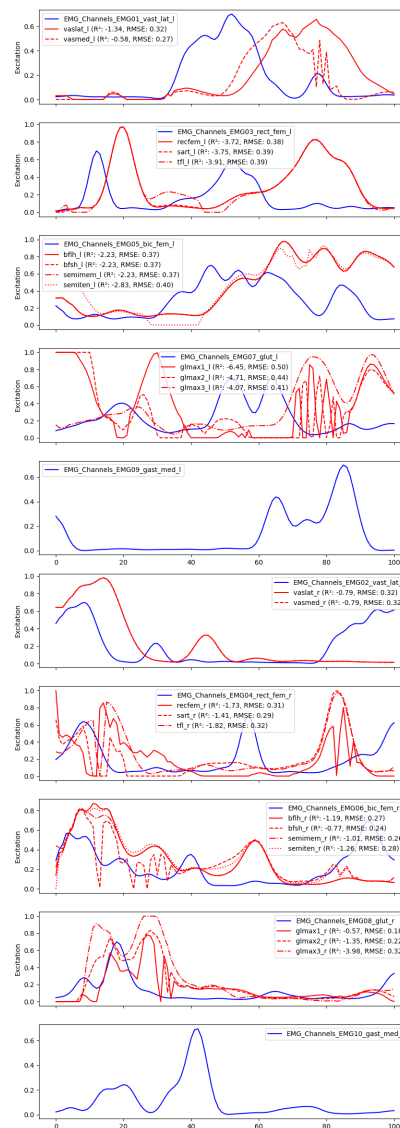


CEINMS
Optimiser
Walking 01

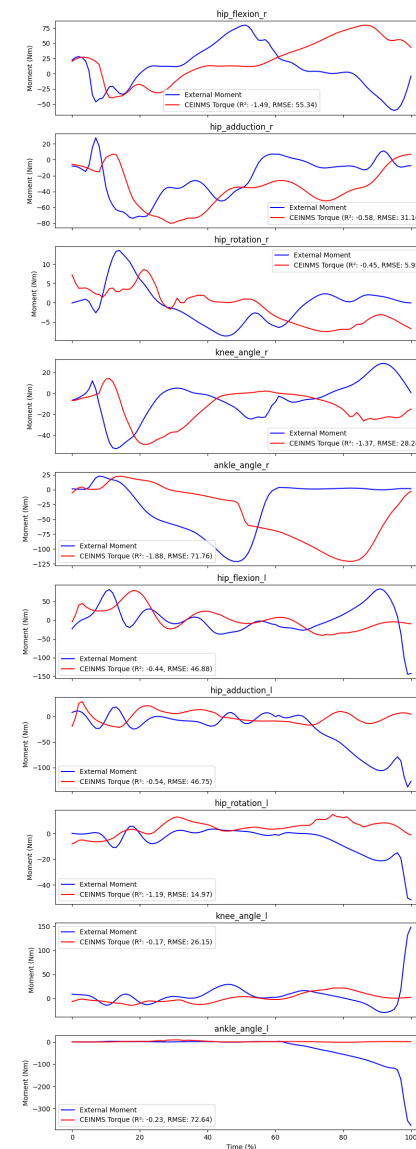
Optimised Values
Alpha = 1
Beta = 2
Gamma = 54

Still Larger error
for final
simulations

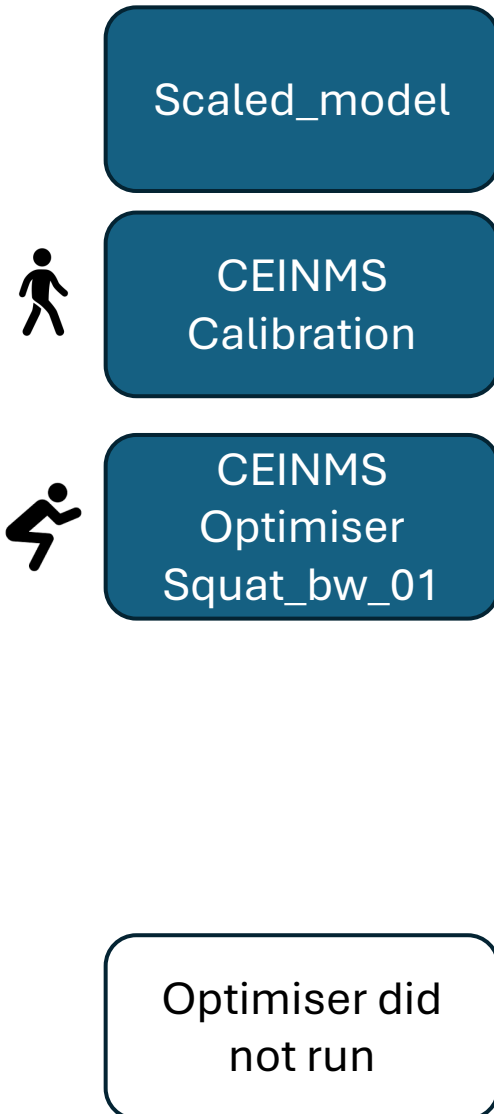
EMG vs CEINMS Excitations



External Torques vs CEINMS Torques



Athlete 03 – session 2 - Squatting



```
perlong_1 -> Synthesising  
piri_1 -> Synthesising  
psoas_1 -> Synthesising  
recfem_1 -> Adjusting  
sart_1 -> Adjusting  
semimem_1 -> Adjusting  
semiten_1 -> Adjusting  
soleus_1 -> Synthesising  
tfl_1 -> Adjusting  
tibant_1 -> Synthesising  
tibpost_1 -> Synthesising  
vasint_1 -> Synthesising  
vaslat_1 -> Adjusting  
vasmed_1 -> Adjusting  
Using alpha = 1, beta = 2, gamma = 5  
Joining thread 132/200  
Joining thread 131/200  
Joining thread 130/200
```

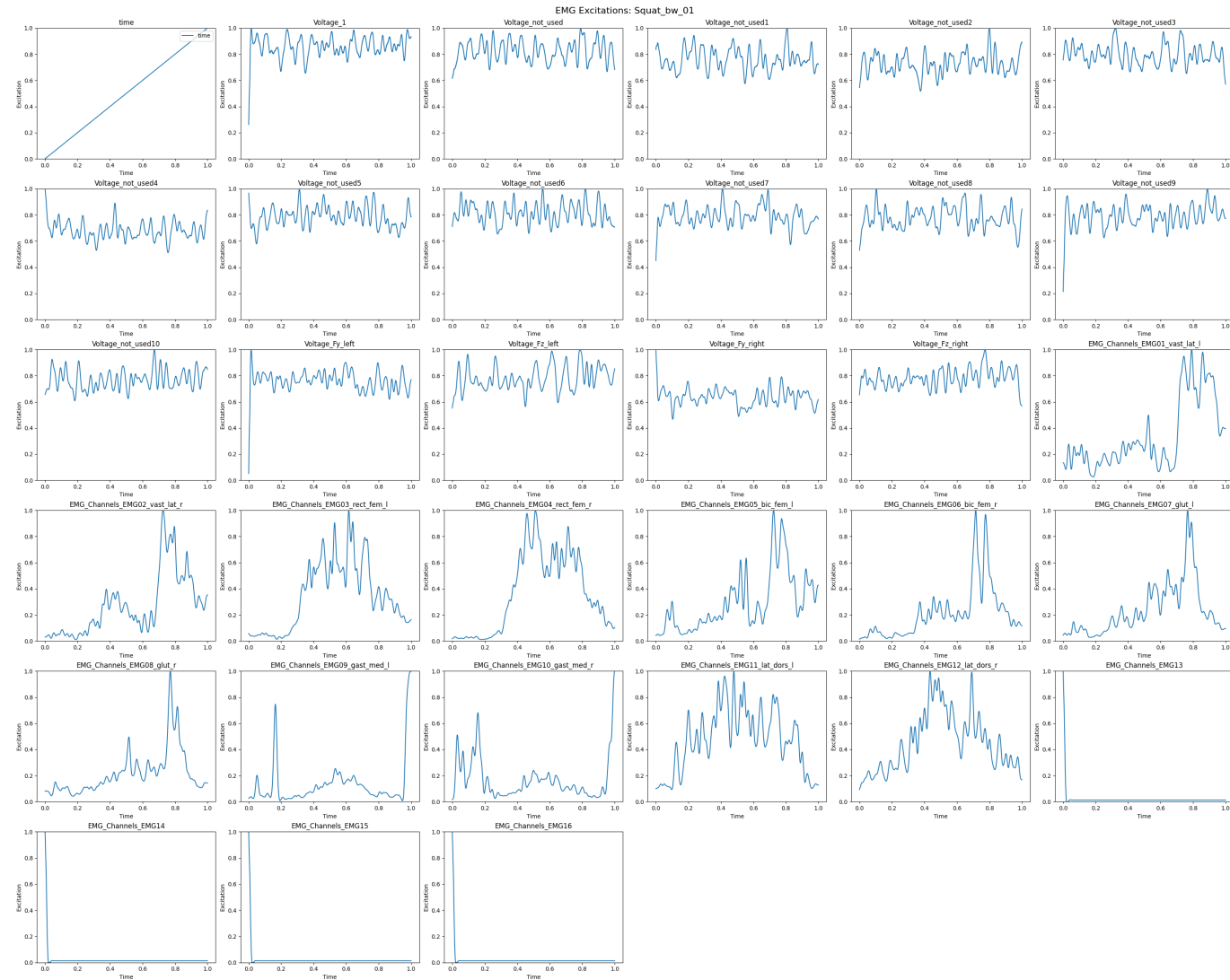
Athlete 03 – session 2 - Squatting

Scaled_model

CEINMS
Calibration

EMG
Squat_bw_01

I assume
problem is the
high EMG
amplitudes



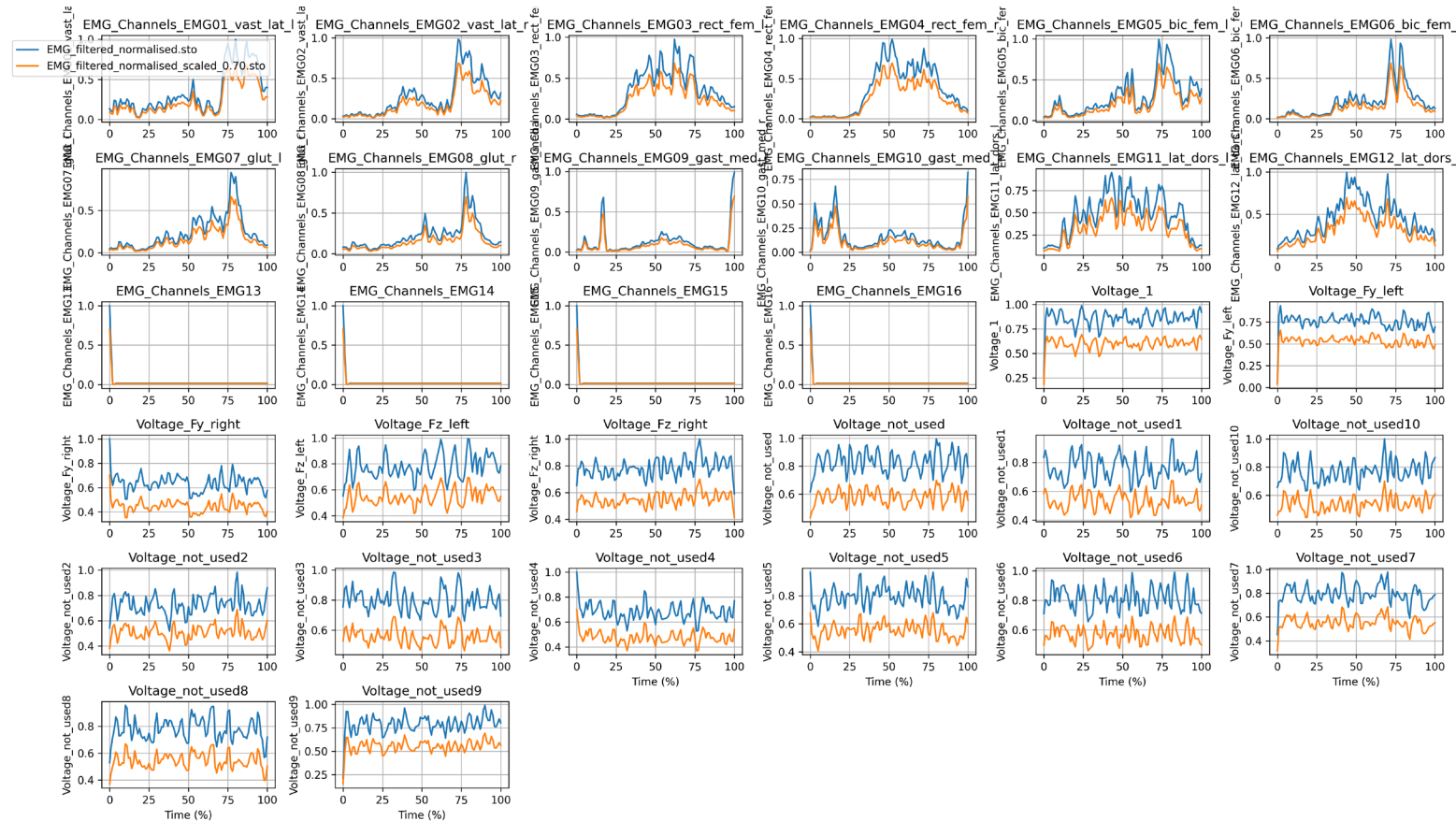
Athlete 03 – session 2 - Squatting

Scaled_model

CEINMS
Calibration

EMG
Squat_bw_01
Factor = 0.7

I assume
problem is the
high EMG
amplitudes



Athlete 03 – session 2 - Walking

Scaled_model



CEINMS
Calibration



CEINMS
Execution
Walking 03

Have you tired?
- Why don't you play around
with Alpha, Beta, Gamma??



Athlete 03 – session 2 - Squatting

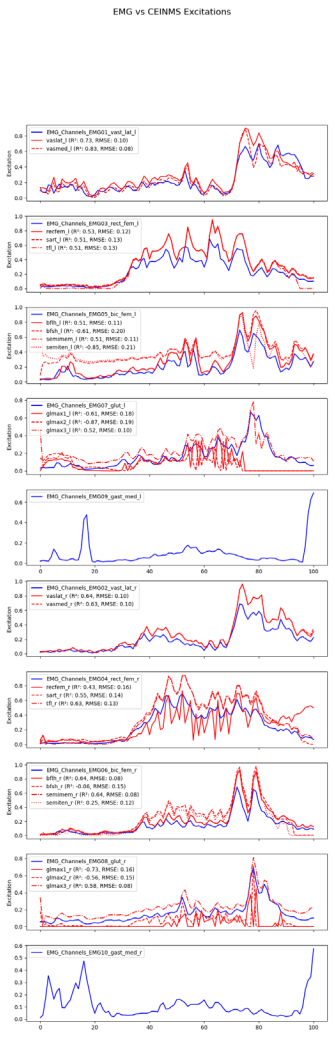
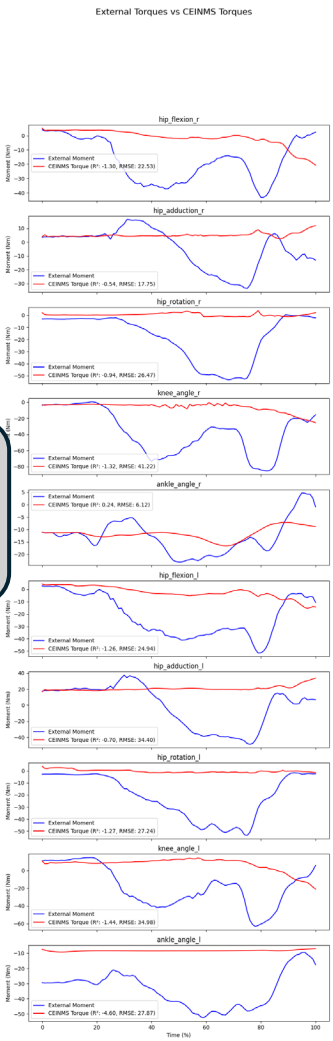
Scaled_model

CEINMS
Calibration

CEINMS
Executions
Squat_bw_01
Factor = 0.7

Optimised Values
Alpha = 1
Beta = 1
Gamma = 34

Large moment
errors



Athlete 03 – session 2 - Squatting

Scaled_model



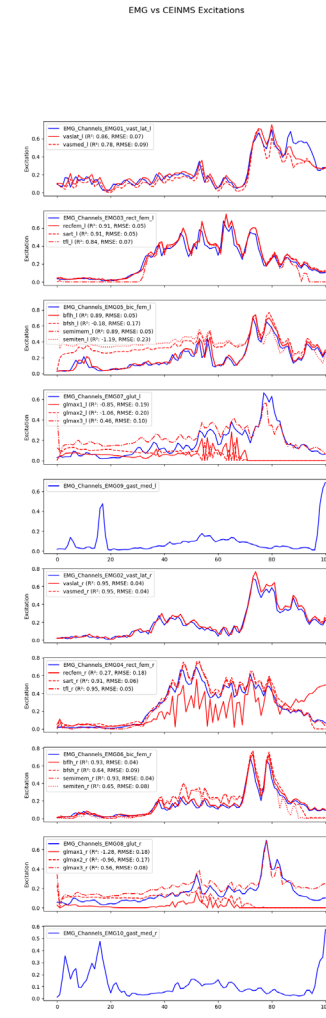
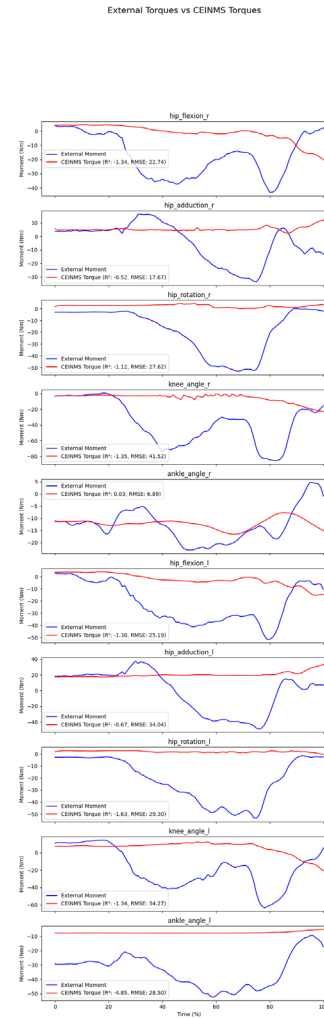
CEINMS
Calibration



CEINMS
Executions
Squat_bw_01
Factor = 0.7

Optimised Values
Alpha = 1
Beta = 10
Gamma = 34

Increased beta
Still Large
moment errors



Athlete 03 – session 2 - Squatting

Scaled_model

CEINMS Calibration

CEINMS
Executions
Squat_bw_01
Factor = 0.7

Optimised Values
Alpha = 1
Beta = 50
Gamma = 34

Still no change
;(
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